PURE MATHEMATICS

REVISION ITEMS

SET 3

DEADLINE: 3 DAYS FROM TODAY.

Section A.

Attempt ALL ITEMS in this section.

Item 1.

At a small welding workshop in Mbale, a technician is designing three rectangular metal sheets for window frames. The areas of the sheets, measured in square centimetres, are expressed in terms of the width x as follows: The first sheet has area $2x^2 + 7x - 4$. The second sheet has area $x^2 + 3x - 4$. The third sheet has area $7x^2 + bx - 8$, where b is a constant yet to be determine. While checking the production plan, the technician notices that all three area expressions have a common linear factor. He must determine this factor before cutting the sheets so that the frames fit perfectly.

Task: Help him determine the common factor of the equations and hence find the value of b

Item 2.

At a grain distribution business in Masindi, a wholesaler packages maize flour in bags whose weights follow a geometric progression. During stock verification, it is observed that the combined weight of the second and third bags is 48 kilograms. Later in the sequence, the combined weight of the fifth and sixth bags is 1296 kilograms. The wholesaler wants to understand the growth pattern of the packaging system in order to plan storage space and transport requirements.

Task: Find the weight of the first bag and hence calculate the total weight of the first 12 bags.

Item 3.

Teddy operates a mobile money business in her trading centre. For security reasons, she trains her assistant never to write her 4-digit Mobile Money PIN anywhere. One day, she tells the assistant that the first three digits are already known to him, but the last digit is hidden in a mathematical condition. She explains that the last digit is the value of x which satisfies the equation $3^x - 3^{2-x} = 8$.

Task: Determine the last digit of Teddy's Mobile Money PIN.

Item 4.

David is installing a solar panel support on a roof. One metal bar makes an angle of 45° with the horizontal, and he knows that $\cos 45° = \frac{\sqrt{2}}{2}$. He fixes another brace exactly halfway between the horizontal and the 45° bar, so it makes an angle of 22.5°. For accurate

measurement, he needs the exact value of $\cos 22.5°$. He recalls that, using trigonometric identities, it can be expressed as $\cos 22.5° = \frac{\sqrt{\sqrt{2}+2}}{2}$ but he has no calculator.

Task:
Help David determine if what he recalls is the exact value of $\cos 22..5°$ using the information given.

Item 5.

Mark, a road construction engineer, models the vertical profile of a new road section using the quadratic expression $y = ax^2 - bx + c$**.** He explains to his team that such a curve can represent either a valley or a hill depending on the values of the constants. From his survey results, he has established that the stationary point of the curve is (2, 4), and that the road crosses the vertical axis at 8 units, meaning the y-intercept is 8. He now needs to determine the exact values of a, b, and c in order to decide whether the road section forms a valley or a hill.

Task: Help Mark determine the values of a, b, and c, and hence establish whether the road section represents a valley or a hill.

Item 6.

A group of computer programmers use the function $f(x) = 2x^2 - 7x - 15$ to control when a certain software feature is activated or deactivated. They explain that the feature switches on when the function gives positive values and switches off when the function gives negative values. They want to know the exact range of values of x for which the function produces positive results and the range for which it produces negative results, but they challenge themselves to determine this mathematically without using a computer.

Task: Help the programmers determine the range of values of x for which the function is;

a) Positive.

b) Negative.

Section B

Attempt ALL ITEMS in this section.

Item 7.

At a regional mathematics contest held in Mbale, finalists were given two independent theoretical investigations. In the first investigation, they were presented with two quadratic expressions: $x^2 + ax + p = 0$ and $cx^2 + 2ax - 3p = 0$. They were informed that the two equations have a common root. Using this condition, they were required to establish the relationship: $p(c + 3)^2 = 5a^2(c - 2)$.

In the second investigation, the contestants analysed a number pattern formed by the series $1^2(1 + 1) + 2^2(2 + 1) + 3^2(3 + 1) + \cdots + n^2(n + 1)$ where n is a whole number. They were

told that this summation can be expressed in the closed form: $\frac{n}{12}(n+1)(n+2)(3n+1)$ and were required to justify this result using the principle of mathematical induction.

Tasks:

a) Using the information given, establish the stated relationship between a, c and p.
b) Prove the stated formula for the summation using mathematical induction.

Item 8.

At a secondary school physics laboratory in Arua, students are conducting two separate investigations during their practical lesson. In the first investigation, a mathematical model describing part of a motion experiment is given by $y = \frac{x^3}{\sqrt{1-2x^2}}$. The students are required to determine how fast y changes with respect to x.

In the second investigation, they study the motion of a simple pendulum. The period T of a pendulum of length l is related by $T^2 = \frac{4\pi^2 l}{g}$ where g is the acceleration due to gravity. During measurement, an error of 2% is made in determining the length l. The students must establish how this measurement error affects the calculated value of the period.

Tasks:

a) Help the students determine how fast y changes with respect to x.
b) Determine the resulting percentage error in the period T when the length l is measured with a 2% error.

Item 9.

During a district mathematics camp in Kabale, senior 6 students were given two independent investigations drawn from trigonometry and geometry. In the first investigation, a model describing the strength of a rotating signal was given as $10\sin x\cos x + 12\cos 2x$. The facilitator informed the students that this expression could be rewritten in the single angle form $R\sin(2x+\alpha)$, where R is positive and α is a constant angle. The students were then asked to use this form to determine the greatest possible value of the original signal expression.

In the second investigation, the students considered a triangular field ABC, where the sides opposite angles A, B and C are denoted by a, b and c respectively. They were reminded that certain relationships between the sides and angles of any triangle allow one to show that $a = b\cos C + c\cos B$ and also that $a\sin\left(\frac{B-C}{2}\right) = (b-c)\cos\left(\frac{A}{2}\right)$. They were required to justify these relationships using appropriate trigonometric identities and properties of triangles.

Tasks:

a) Rewrite the given signal expression in the required single angle form and hence determine its maximum value.
b) Prove the two stated relationships for triangle ABC.

Item 10.

At a technology innovation hub in Kampala, a group of senior students is developing a software module that handles complex number computations for signal processing. In testing the program, they are required to simplify the expression:

$(2+5i)^2 + 5\frac{7+2i}{3-4i} - i(4-6i)$ and present the final result in the standard form $a+bi$.

In another part of the project, the students are modelling positions on a digital Argand diagram. A complex number is represented by $z = x + yi$, where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. They are informed that the locus of points satisfying $Arg\left(\frac{z-1}{z-i}\right) = \frac{\pi}{3}$ forms a circle in the complex plane. They must determine the geometric features of this locus.

Tasks:

a) Simplify the given complex number expression and express the result in the form $a + bi$.
b) Show that the stated locus represents a circle and determine its centre and radius.

Item 11.

At a municipal land office in Entebbe, a surveyor is analysing two different property layouts submitted for approval. In the first layout, a triangular plot of land is enclosed by three straight boundary lines given by: $x + y = 11, y = x - 1$ and $3y = x - 3$. The surveyor must determine the exact balance point of the triangular plot for the purpose of installing a central water outlet.

In the second layout, a developer proposes a rhombus-shaped recreational park labelled ABCD. Two opposite vertices are given as A(−3, −4) and C(5, 4). It is further specified that the gradient of side BC is 2. The surveyor needs to determine the equation of the diagonal BD, establish the remaining vertices B and D, and confirm the total area of the park before approval.

Tasks:

a) Determine the centroid of the triangular plot formed by the three given boundary lines.
b) Find the equation of the diagonal BD of the rhombus. Hence determine the coordinates of B and D and show that the area of the rhombus is $21\frac{1}{3}$ square units.

Item 12.

At a technical training institute in Hoima, students are carrying out two independent applied mathematics investigations. In the first investigation, a moving particle has its position defined parametrically by $x = \frac{3t-1}{t}$ and $y = \frac{t^2+4}{t}$ where t is a parameter. The students are required to analyse the curvature of the path by determining the second derivative of y with respect to x, and to show that $\frac{d^2y}{dx^2} = 2t^3$.

In the second investigation, the institute plans to fence off a rectangular demonstration garden using 100 metres of fencing wire. One side of the rectangle will lie along an already existing straight wall, so fencing is required for only three sides. The management wants to maximise the enclosed area using the available fencing.

Tasks:

a) Using the given parametric equations, show that the second derivative of y with respect to x is $2t^3$.
b) Determine the maximum possible area of the rectangular garden that can be enclosed under the given conditions.

Item 13.

At a roadside food stall in Soroti, a vendor prepares a large flask of tea early in the morning. After removing the flask from the charcoal stove, the vendor notices that the tea begins to cool in such a way that the rate at which its temperature falls is proportional to the difference between the temperature of the tea and that of the surrounding air. At the moment the flask is removed from the stove, the temperature of the tea is 60°C. After 15 minutes, the temperature has dropped to 50°C. The surrounding air temperature remains constant at 10°C throughout the cooling process. The vendor wants to predict how long it will take before the tea cools to a comfortable serving temperature of 30°C.

Tasks:

a) Form a differential equation to model the temperature of the tea at any time (t) minutes after removal from the stove.

b) Determine the time it takes for the temperature of the tea to reach 30°C.

Item 14.

During a mathematics innovation workshop at your school, a group of Senior Four learners are using graphing software to model the behaviour of a machine component. The displacement y of the component is related to a control variable x by the equation $y = x^3 - 6x^2 + 9x + 2$. When analysing the performance of the component, the learners are particularly interested in the point where the displacement becomes zero. From their plotted graph, they observe that one of the points where the curve crosses the horizontal axis lies between -1 and 0.

To improve the accuracy of their estimate, the mathematics club patron introduces an iterative method. After differentiating the model and substituting into the Newton–Raphson procedure, the updating formula for successive approximations is obtained as

$x_{n+1} = \frac{2}{3}\left[\frac{x^3{}_n - 3x^2{}_n - 1}{x^2{}_n - 4x_n + 3}\right]$ where $n = 0,1,2,3$ ... The learners decide to begin with the graphical estimate of the root between -1 and 0 as their initial approximation.

Task:

a) Find graphically the root of the equation that lies between -1 and 0.
b) i) Show that applying the Newton–Raphson method to the given function leads to the iterative formula.
   ii) Using the iterative formula and the initial approximation obtained in (a), determine the root correct to two decimal places.

Item 15.

An engineer operating a small fruit-processing factory in Namanve Industrial Area is studying the performance of a new automated juice-mixing machine. During calibration, the relationship between the machine's control setting x (measured in radians) and the output concentration level y is found to satisfy $x^2 \sec x - xy + 2y^2 = 15$ where y is a differentiable function of x.

The machine is adjusted using a rotating control knob. If the angle of rotation is θ, the system is modelled by $x = \sin\theta$ and $y = \theta - \cos\theta$**.** Further testing shows that the production efficiency E, depending on the control setting x, is given by E= x^2e^{-x}. One afternoon, the engineer observes that when the control setting is $x = \frac{\pi}{3}$, the output concentration level is $y = 2$.

Tasks:

a) Determine the rate of change of the output concentration level with respect to the control setting when $x = \frac{\pi}{3}$ and $y = 2$.
b) Show that the second derivative of y with respect to x in terms of θ is $\frac{d^2y}{dx^2} = \frac{1+sin\theta}{cos^3\theta}$.
c) Determine the control setting that gives maximum production efficiency and hence find the maximum efficiency.

END

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